SERIES-C

UNIVERSITY ENTRANCE TEST FOR ADMISSION TO UNDER-GRADUATE PROGRAMMES-2020

SERIES **C** | **Booklet No:**

Time Allowed : 2 hours 15 minutes for PCB &PCM and 3hours for PCMB

This booklet contains 180 questions and 19 Printed pages
Do not open this booklet until you are asked by the invigilator.

Important Note:

The candidates will solve the questions of the subjects of their streams only e.g. PCB candidates shall attempt Q.1-135, PCM candidates Q.1-90 & Q.136-180 and PCBM candidates shall attempt all 180 Questions.

Instructions:

1. Immediately after opening the Question Booklet, ensure that it contains the number of pages & Questions as written above. In case of any discrepancy, report to the invigilator concerned.
2. Use only blue/black ball-pen for writing your particulars like Name, Roll No. Question Booklet No. on the OMR answer sheet and two digit subject code (I.e 01, 02 & 03) at spaces provided.
3. OMR answer sheets without mentioning Question Booklet number will not be evaluated.
4. Use only blue/black ball-pen for writing your most appropriate responses on the OMR answer sheet.
5. Mark only the responses to the questions on OMR sheets. Erasing cutting or overwriting on OMR Sheets will be considered to have not been attempted.
6. The candidates should on demand show their admit cards to invigilators or any other authorized representative of University
7. No candidate will be allowed to leave the examination hall before half time except under special circumstances with the permission by Centre Superintendent.
8. The candidates shall hand over the filled OMR answer sheet to invigilator before leaving the examination hall.
9. The candidate is NOT allowed to carry any stationary item like textual material (printed/written), bits of papers, geometry/ pencil box, plastic pouch, calculator, writing pad, Pen Drives, Erasers, Log Table, Electronic Pen/ Scanner etc.
10. The candidate is NOT allowed to carry any communication device like Mobile Phone, Bluetooth, Earphones, Microphone, Pager, Health Band, Smart-Watch, Camera etc. Other items like Wallet, Googles, Handbags, Belt, Cap, ATM/Credit/Debit Card, Plastic identity card, any remote key etc.
11. The Candidates are not allowed to use white correction fluid on the OMR sheet & Attendance Sheet.
12. The Candidate should sit on their allotted seats only.
13. Non-adherence to above would be considered as cases of Unfair-means.
14. The cases pertaining to the use of unfair means will be governed by the rules and regulations of University.

PHYSICS 1-45 TO BE ATTEMPTED BY CANDIDATES OF ALL STREAMS

Q1 The bulk modulus of spherical object 'B'. If it is subjected to uniform pressure 'p', the fractional decrease in radius is

(A) p/B (B) B/3p

(C) 3p/B (D) p/3B

Q2 A small sphere of radius 'r' falls from rest in a viscous liquid. As a result heat produced due to viscous force. The rate of production of heat when the sphere attains its terminal velocity, is proportional to

(A) r^5 (B) r^2

(C) r^3 (D) r^4

Q3 An ideal gas is expanded such that PT^2 =a constant. The coefficient of volume expansion of the gas is

$PT^2 = a$

(A) 1/T (B) 2/T

(C) 3/T (D) 4/T

Q4 If C_p and C_v denote the specific heats (per unit mass of an ideal gas of molecular weight M), then

$C_p - C_v = R$

(A) $C_p - C_v = R/M^2$ (B) $C_p - C_v = R$

(C) $C_p - C_v = R/M$ (D) $C_p - C_v = MR$

Q5 Two particles are oscillating along two close parallel straight lines side by side with the same frequency and amplitude. They pass each other, moving in opposite direction when their displacement is half of the amplitude. The mean positions of two particles lie on a straight line perpendicular to the paths of the two particles. The phase difference is

(A) 0 (B) $2\pi/3$

(C) π (D) $\pi/6$

Q6 The equation of simple harmonic wave is given by y= 3 Sin π/2 (50t – x) where x and y in meters and t in seconds. The ratio of maximum particle velocity to the wave velocity is

(A) 2π (B) $3/2\ \pi$

(C) 3π (D) $2/3\ \pi$

Q7 When 10^{19} electrons are removed from a neutral metal plate through some process, then the charge on it becomes :

(A) +1.6 C (B) - 1.6 C

(C) 10^{19} C (D) 10^{-19} C

Q8 Two equal and opposite charges of masses m_1 and m_2 are accelerated in a uniform electric field through the same distance. What is the ratio of their accelerations, if their ratio of masses is $m_1/m_2 = 0.5$?

$\frac{m_1}{m_2} = 0.5$

(A) $a_1/a_2 = 0.5$ (B) $a_1/a_2 = 1$

(C) $a_1/a_2 = 2$ (D) $a_1/a_2 = 3$

Q9 Equipotentials at a great distance from a collection of charges whose total sum is not zero are approximately:

(A) spheres (B) planes

(C) paraboids (D) ellipsoids

Q10 A positive charge is released from rest in an uniform electric field. The electric potential energy of the charge:

(A) remains a constant because the electric field is uniform.

(B) increases because the charge moves along the electric field.

(C) decreases because the charge moves along the electric field.

(D) decreases because the charge moves opposite to the electric field.

Q11 The supply voltage to room is 120 V. The resistance of lead wire is 6Ω. A 60 W bulb is already switched on. What is the decrease of voltage across the bulb, when a 240 W heater is switched on in parallel to the bulb ?

(A) zero volt (B) 2.9 V

(C) 13.3 V (D) 10.04 V

Q12 When 5 V potential difference is applied across a wire of length 0.1 m the drift speed of electrons is 2.5×10^{-4} m/s. If the electron density in the wire is $8\times 10^{28}\ m^{-3}$, the resistivity of the material is close to:

(A) $1.6 \times 10^{-8}\Omega m$ (B) $1.6 \times 10^{-7}\Omega m$

(C) $1.6 \times 10^{-6}\Omega m$ (D) $1.6 \times 10^{-5}\Omega m$

Q13 Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above the other such that O is their common point for the two.The wires carry I_1 and I_2 currents respectively. Point P is lying at distance d from O along a direction perpendicular to the plane containing the wires. The magnetic field at point P will be:

(A) $\frac{\mu_0}{2\pi d}\ \frac{I_1}{I_2}$ (B) $\frac{\mu_0}{2\pi d}\ (I_1 - I_2)$

(C) $\frac{\mu_0}{2\pi d}\ (I_1^2 - I_2^2)$ (D) $\frac{\mu_0}{2\pi d}\ (I_1^2 + I_2^2)^{1/2}$

Q14 Two identical wires A and B each of length L carry the same current I. Wire A is bent into a circle of radius R and wire B is bent to form a square of side a. If B_A and B_B are the values of magnetic field at the centre of the circle and square respectively, then B_A / B_B is :

(A) $\pi^2/8$ (B) $\pi^2/16\sqrt{2}$

(C) $\pi^2/16$ (D) $\pi^2/8\ \sqrt{2}$

Q15 When a current of 5mA is passed through a galvanometer having a coil of resistance 15Ω, it shows full scale deflection. The value of the resistance to be put in series with the galvanometer to convert it into voltmeter of range 0- 10V is:

(A) $2.535 \times 10^3\Omega$ (B) $4.005 \times 10^3\Omega$

(C) $1.985 \times 10^3\Omega$ (D) $2.045\times 10^3\Omega$

Q16 The coercivity of a small magnet where the ferromagnet gets demagnetized is $3 \times 10^3\ Am^{-1}$. The current required to be passed in a solenoid of length 10 cm and number of turns 100,so that the magnet get demagnetized when inside the solenoid is:

(A) 3 A (B) 6 A

(C) 30 mA (D) 60 mA x

Q17 A 50 Hz AC current of crest value 1A flows through the primary of a transformer. If the mutual inductance between the primary and secondary be 0.5H, the crest voltage inductance in the secondary is:

(A) 75 V (B) 150 V

(C) 100 V (D) 200 V

Q18 A transformer having efficiency of 90% is working on 200 V and 3 KW power supply. If the

$\frac{N_S I_S}{N_P I_P} =$ $\eta = \frac{V_S I_S}{V_P I_P}$ $\frac{90}{100}$ $0.9 =$ x6 current in secondary coil is 6A, the voltage across the secondary coil and the current in the primary coil respectively are:

(A) 300V, 15A (B) 450V, 15A
(C) 450V, 135A (D) 600V, 15A

Q19 An inductor 20mH, a capacitor 50μF and a resistor 40Ω are connected in series across a source of e.m.f. V = 10 sin340t. The power loss in a.c. circuit is:

(A) 0.67W (B) 0.76W
(C) 0.89W (D) 0.046W

Q20 Arrange the following electromagnetic radiations per quantum in order of increasing energy:

a: Blue light b: Yellow light c: X-rays d: Radiowave

(A) d, b,a,c (B) a,b,d,c
(C) c,a,b,d (D) b,a,d,c

Q21 A source of light lies on the angle bisector of two plane mirrors inclined at an angle θ. The value of θ, so that the light reflected from one mirror does not reach the other mirror will be:

(A) $\theta \geq 120^\circ$ (B) $\theta \geq 90^\circ$
(C) $\theta \leq 120^\circ$ (D) none of the above

Q22 A car is fitted with a convex side- view mirror of focal length 20cm. A second car 2.8m behind the first car is over taking the first car at relative speed of 15m/s.The speed of the image of the second car as seen in the mirror of the first one is: f = 20cm. 15 m/sec.

(A) 1/15 m/s (B) 10m/s
(C) 15m/s (D) 1/10 m/s

Q23 A diverging lens with magnitude of focal length 25 cm is placed at a distance of 15 cm from a converging lens of magnitude of focal length 20 cm. A beam of parallel light falls on the diverging lens. The final image formed is :

(A) real and at a distance of 40 cm from the diverging lens.
(B) real and at a distance of 6 cm from the converging lens.
(C) real and at a distance of 40 cm from the convergent lens.
(D) virtual and at a distance of 40 cm from the convergent lens.

Q24 A thin prism of angle 15° made of glass of refractive index $\mu_1 = 1.5$ is combined with another prism of glass of refractive index $\mu_2 = 1.75$. The combination of prism produces dispersion without deviation. The angle of the second prism should be:

(A) 7° (B) 10°
(C) 12° (D) 5°

Q25 Young's double slit experiment is first performed in air and then in a medium other then air. It is found that 8th bright fringe in the medium lies where the 5th dark fringe lies in the air. The refractive index of the medium is nearly:

(A) 1.25 (B) 1.59
(C) 1.69 (D) 1.78

Q26 A beam of unpolarised light of intensity I_0 passes through a polariod A and then through another polariod B which is oriented so that its principal plane makes an angle of 45° relative to that of A. The intensity of emergent light is:

(A) I_0 (B) $I_0/2$
(C) $I_0/4$ (D) $I_0/8$

Q27 When the energy of incident radiation is increased by 20%, the kinetic energy of photoelectrons emitted from a metal surface is increased from 0.5 eV to 0.8 eV. The work function of metal is:

·(A) 0.65 eV
(B) 1.0 eV
(C) 1.3 eV
(D) 1.5 eV

Q28 The De-Broglie wavelength of a particle moving with a velocity of 2.25×10^8 m/s is equal to wavelength of photon. The ratio of kinetic energy of the particle to the energy of photon (velocity of light is 3×10^8 m/s):
(A) 1/8
·(B) 3/8
(C) 5/8
(D) 7/8

Q29 Hydrogen atom in ground state is excited by a monochromatic radiation of λ = 975 A°. Number of special lines in the resulting spectrum emitted will be:
$\frac{K}{E} = \frac{v}{2c}$
(A) 3
·(B) 2
(C) 6
(D) 10

Q 30 Radioactive material 'A' has decay constant '8λ' and material 'B' has decay constant 'λ'. Initially they have same number of nuclei. After what time the ratio of number of nuclei of material B to that A' will be $1/e$:
$\frac{K}{E} = \frac{2.25\times10^8}{2\times3\times10^8}$
·(A) $1/\lambda$
(B) $1/7\lambda$
(C) $1/8\lambda$
(D) $1/9\lambda$

Q31 The binding energy per nucleon of $_3Li^7$ and $_2He^4$ nuclei are 5.60 MeV and 7.06 MeV respectively. In the nucleon reaction $_3Li^7 + _1H^1 \rightarrow _2He^4 + _2He^4 + Q$, the value of energy Q released is:
$\frac{K}{E} = \frac{2.25}{6}$
·(A) 19.6 MeV
(B) - 2.4 MeV
(C) 8.4 MeV
(D) 17.3 MeV

Q32 In p-n junction, the barrier potential offers resistance to;
(A) only free electrons in n-region
(B) only holes in p-junction
✓(C) free electrons in n-region and holes in p- region
(D) free electrons in p- region and holes in n- region

Q 33 The manifestation of band structure in solid is due to:
(A) Heisenberg uncertainty principle
(B) Pauli's exclusion principle
(C) Bohr's correspondence principle
·(D) Boltzmann law

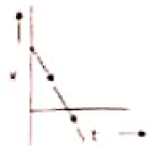
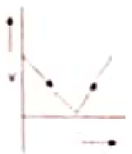
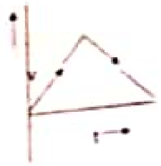
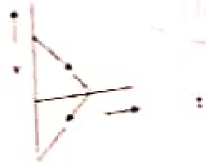
Q34 If force ('F'), velocity (V) and time (T) are taken as fundamental units, the dimensions of mass are
(A) $[FVT^{-1}]$
·(B) $[FVT^{-2}]$
(C) $[FV^{-1}T^{-1}]$
(D) $[FV^{-1}T]$

Q35 A particle moves in XY plane in such a way that its x and y co-ordinates vary with time according to
$x(t) = t^3 - 32t$
$y(t) = 5t^2 + 12$
find the acceleration of the particle, if t= 3 sec.
$\frac{K}{E} = \frac{225}{60} \frac{45}{12} \frac{15}{4}$
·(A) 9i+5j
(B) 18i+10j
(C) 18i-5j
(D) -18i+10j

Q36 A body of mass m moving along a straight line covers half the distance with a speed of 2m/s. The remaining half of the distance is covered in two equal time intervals with a speed of 3m/s and 5m/s respectively, the average speed of the particle for the entire journey is
$3^3 - 32(3)$
$27 - 96$
(A) (3/8) m/s
·(B) (8/3) m/s
(C) (4/3) m/s
(D) (16/3) m/s

λ particle = λ photon
$\frac{h}{mc} = \frac{hc}{E}$
$\frac{K}{E} = \frac{v}{\lambda c}$

Q37 A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity verses time?

(A)  (B)  (C)  (D) 

Q38 For a particle in uniform circular motion, the acceleration a at a point P(R, θ) on the circle of radius R is (Here θ is measured from the x-axis)

(A) $-v^2/R\ \sin\theta\ i + v^2/R\ \cos\theta\ j$ (B) $-v^2/R\ \cos\theta\ i - v^2/R\ \sin\theta\ j$

(C) $v^2/R\ i + v^2/R\ j$ (D) $-v^2/R\ \cos\theta\ i - v^2/R\ \sin\theta\ j$

Q39 A stone is dropped from a height h. It hits the ground with a certain momentum p. If the same stone is dropped from a height 100% more than the previous height, the momentum when it hits the ground will change by:

(A) 68% (B) 41 %

(C) 200% (D) 100%

Q40 A body of mass 40 kg resting on rough horizontal surface is subjected to a force P which is just enough to start the motion of the body, If $\mu_s = 0.5$, $\mu_k = 0.4$, $g = 10 m/s^2$ and the force P is continuously applied on the body, then the acceleration of the body is

(A) zero (B) $1 m/s^2$

(C) $2 m/s^2$ (D) $2.4 m/s^2$

Q41 In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50% greater than the original kinetic energy, the magnitude of the relative velocity between the two particles after collision is

$a = \frac{Af - Lu}{2}$

(A) $v_0/4$ (B) $\sqrt{2}\, v_0$

(C) $v_0/2$ (D) $v_0/\sqrt{2}$

Q42 Distance of the centre of mass of a solid uniform cone from its vertex is Z_0. If the radius of its base is R and its height h, then Z_0 is equal to

$a = \frac{0.5 - 0.4}{2}$

(A) $h^2/4R$ (B) $3h/4$

(C) $5h/8$ (D) $3h^2/8R$

Q43 A disc and a solid sphere of same radius but different masses roll off on two inclined planes of the same altitude and length, which one of the two objects gets to the bottom of the plane first?

$a = \frac{0.1}{2}$

(A) solid sphere (B) both reach at the same time

(C) depends on their masses (D) disc

Q44 A solid sphere is rotating freely about its symmetry axis in free space. The radius of the sphere is increased keeping its mass same, which of the following physical quantities would remain constant for the sphere ?

$a = \frac{1}{20}$

(A) Rotational kinetic energy (B) Moment of Inertia

(C) Angular velocity (D) Angular momentum

Q45 If the mass of the sun were ten times smaller and the universal gravitational constant were 10 times larger in magnitude, which of the following is not correct?

(A) Time period of simple pendulum on Earth would decrease.

(B) walking on ground would become more difficult.

20 | 100 | 0.05

$a = \frac{Af - Lu}{2} = \frac{0.5 - 0.4}{2}$

$\frac{0.1}{2}$

$\frac{1}{2}$

$\frac{1}{20}$

CHEMISTRY (Q 46-90 TO BE ATTEMPTED BY CANDIDATES OF ALL STREAMS

Q 46 $CuCO_3 \xrightarrow{\Delta} (R) + CO_2$. Oxides (R) shows

(A) Acidic nature (B) Basic nature

(C) Neutral (D) Amphoteric

Q 47 XY_2 dissociates as – $XY_2 \rightleftharpoons XY(g) + Y(g)$. Initial pressure of XY_2 is 600 mm Hg. The total pressure at equilibrium is 800 mm Hg. Assuming volume of system to remain constant, the value of K_p is

(A) 50 (B) 100

(C) 20 (D) 400

Q 48 Maximum number of hydrogen bonds formed by water molecule H_2O

H – O – H

(A) 1 (B) 2

(C) 3 (D) 4

Q 49 Which one of the following crystals is represented by $a \neq b \neq c$ and $\alpha \neq \beta \neq \gamma$

(A) Orthorhombic (B) Monoclinic

(C) Triclinic (D) Tetragonal

Q 50 The correct IUPAC name of $Fe(C_5H_5)_2$ is

(A) Cyclopentadienyliron (B) Bis(Cyclopentadienyl)iron

(C) Dicyclopentadinyl Ferrate(II) (D) Ferrocane

Q 51 Which of the following is an example of semi-synthetic polymer?

(A) Protein (B) Plastic

(C) Vulcanised rubber (D) Cellulose

Q 52 The number of octahedral voids per atom present in a cubic close packed structure is

(A) 1 (B) 3

(C) 2 (D) 4

Q 53 Metal X does not react with cold water but evolves highly inflammable gas with steam. Then X can be

(A) Na (B) Fe

(C) Ni (D) Cr

Q 54 6.02×10^{20} molecules of urea are present in 100ml of its solution. The concentration of solution is

(A) 0.01 M (B) 0.001 M

(C) 0.1 M (D) 0.02 M

Q 55 Ice and water kept in a perfectly insulated thermos flask (no exchange of heat between its contents and the surroundings) at 273K and the atmospheric pressure are in equilibrium state and the system shows

(A) Liquid - liquid equilibrium (B) Liquid - vapour equilibrium

(C) Solid - liquid equilibrium (D) Solid - gas equilibrium

Q 56 Electrolytic refining is used to purify which of the fallowing metals

(A) Cu and Zn (B) Ge and Si

(C) Zr and Ti (D) Zn and Hg

Q 57 Bessemer converter is used for the preparation of

(A) Steel (B) Wrought iron

(C) Pig iron (D) Cast iron

Q.58 Equal volumes of two gases which do not react together are enclosed in separate vessels. Their pressures at 100 *mm* and 400 *mm* respectively. If the two vessels are joined together, then what will be the pressure of the resulting mixture (temperature remaining constant)

(A) 125 *mm* (B) 500 *mm*

(C) 1000 *mm* (D) 250 *mm*

Q.59 According to Bronsted-Lowry concept, the correct order of relative strength of bases follows the order

(A) $CH_3COO^- > Cl^- > OH^-$ (B) $CH_3COO^- > OH^- > Cl^-$

(C) $OH^- > CH_3COO^- > Cl^-$ (D) $OH^- > Cl^- > CH_3COO^-$

Q.60 Atomic radii of alkali metals (M) follow the order : Li < Na < K <Rb but ionic radii in aqueous solution follow the reverse order Li+ > Na+ > K+ >Rb+. The reason for the reverse order is -

(A) Increase in the ionisation energy

(B) Decrease in the metallic bond character

(C) Increase in the electropositive character

(D) Decrease in the amount of hydration

Q.61 Which one of the following is expected to exhibit optical isomerism (*en* = ethylenediamine)

(A) $cis-[Pt(NH_3)_2Cl_2]$ (B) $trans-[Co(en)_2Cl_2]$

(C) $trans-[Pt(NH_3)_2Cl_2]$ (D) $cis-[Co(en)_2Cl_2]$

Q.62 Both Be and Al become passive on reaction with conc. nitric acid due to –

(A) The non reactive nature of the metal (B) The non reactive nature of the acid

(C) The formation of an inert oxide layer on the surface of the metals

(D) None of these

Q.63 Which of the following can be used as the halide component for Friedel Crafts reaction?

(A) Chlorobenzene (B) Bromobenzene

(C) Chloroethene (D) Isopropyl chloride

Q.64 Boron has an extremely high melting point because of

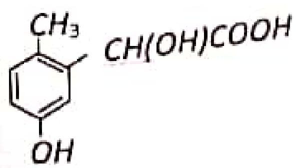
(A) the strong van der Waals forces between its atoms

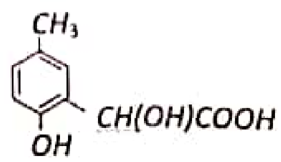
(B) the strong binding forces in the covalent polymer

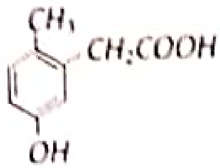
(C) its ionic crystal structure

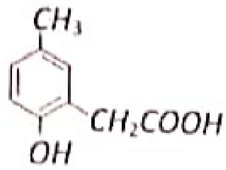
(D) allotropy

Q.65 p-cresol reacts with chloroform in alkaline medium to give the compound A which adds hydrogen cyanide to form, the compound B. The latter on acidic hydrolysis gives chiral carboxylic acid The structure of the carboxylic acid is

(A) 

(B) 

(C) 

(D) 

Q66. The correct set of four quantum numbers for the outermost electron of Na (Z=11) is

(A) n=3, l=0, m=0, s=1/2

(B) n=3, l=1, m=1, s=1/2

(C) n=3, l=0, m=1, s=1/2

(D) n=4, l=0, m=0, s=1/2

Q67. $CaC_2 + H_2O \rightarrow A \xrightarrow{H_2SO_4,\ HgSO_4} B$. Identify *A* and *B* in the given reaction

(A) C_2H_2 and CH_3CHO

(B) CH_4 and $HCOOH$

(C) C_2H_4 and CH_3COOH

(D) C_2H_2 and CH_3COOH

Q68. A group which deactivates the benzene ring towards electrophilic substitution but which directs the incoming group principally to the *o*- and *p*-positions is

(A) $-NH_2$

(B) $-Cl$

(C) $-NO_2$

(D) $-C_2H_5$

Na = 11
$1s^2, 2s^2, 2p^6, 3s^1$
n=3,

Q69. Identify Z in the following series

$$CH_2 = CH_2 \xrightarrow{HBr} X \xrightarrow{Hydrolysis} Y \xrightarrow[I_2\ excess]{Na_2CO_3} Z$$

(A) C_2H_5I

(B) C_2H_5OH

(C) CHI_3

(D) CH_3CHO

Q70. KI and copper(II) sulphate mixed to give

(A) CuI_2 + K2SO4

(B) $Cu_2I_2 + K_2SO_4$

(C) $K_2SO_4 + Cu_2I_2 + I_2$

(D) $K_2SO_4 + CuI_2 + I_2$

Q71. Which of the following acids on decarboxylation gives isobutane

(A) 2,2–Dimethyl butanoic acid

(B) 2,2–dimethyl propanoic acid

(C) 3–Methyl pentanoic acid

(D) 2–Methyl butanoic acid

Q72. The emperical formula of compound is CH_2O. If its molecular weight is 180. The molecular formula of the compound is

(A) $C_3H_6O_3$

(B) $C_4H_8O_4$

(C) $C_6H_{12}O_6$

(D) $C_5H_{10}O_5$

Q73. The sharp melting point of crystalline solids is due to ...A... Here, A refers to

(A) A regular arrangement of constituent particles observed over a short distance in the crystal lattice

(B) A regular arrangement of constituent particles observed over a long distance in the crystal lattice

(C) Same arrangement of constituent particles in different directions

(D) Different arrangement of constituent particles in different directions

Q.74 Which of the following does not give yellow precipitate with I_2 and $NaOH$

(A) C_2H_5OH (B) CH_3CHO

(C) CH_3COCH_3 (D) $HCHO$

Q.75 Which of the fallowing is not correct?

A Milk is a naturally occurring emulsion

B Gold sol is a lyophilic sol

C Physical adsorption decreases with rise in temperature

D Chemical adsorption is unilayered

Q.76 Which one of the following statements about starch is correct?

(A) It occurs in the cell walls of plants

(B) It is a disaccharide

(C) It gives a dark blue colour with iodine solution

(D) It gives a red orange precipitate on boiling with Fehling's solution

Q.77 Terylene is

(A) An addition polymer with a benzene ring in every repeating unit

(B) A condensation polymer with benzene ring in every repeating unit

(C) An addition polymer with two carbon atoms in every repeating unit

(D) A condensation polymer with two nitrogen atoms in every repeating unit

Q.78 A nitrogen containing organic compound on heating with chloroform and alcoholic *KOH*, evolved very unpleasant smelling vapour. The compound could be

(A) *N, N*-dimethyl amine (B) Nitrobenzene

(C) Aniline (D) Benzamide

Q.79 Modern periodicity law is :-

(A) The physical and chemical properties of the elements are periodic function of their atomic number

(B) The physical and chemical properties of the elements depend upon the energy of the electron

(C) The physical and chemical properties of the elements depend upon atomic weight

(D) None of these

Q.80 Which of the following is incorrect for an ideal solution :-

(A) $\Delta H_{mix} = 0$ (B) $\Delta V_{mix} = 0$

(C) $\Delta P = P_{obs} - P_{calculated} = 0$ (D) $\Delta G_{mix} = 0$

Q.81 99% of a first order reaction was completed in 32 minutes. When will 99.9% of reaction complete?

(A) 50 minutes (B) 46 minutes

(C) 49 minutes (D) 48 minutes

Q.82 Baeyer's reagent is:-

(A) 1% alkaline $KMNO_4$ (B) Acidic $KMNO_4$

(C) Neutral $KMNO_4$ (D) Aqueous Br Solution

Q.83 The normality of 1M solution of H_3PO_4 is

$t_{99\%} = 2 \times 32 = 64$

$t_{99.9\%} = 3 \times 32 = 96$

$N = 2 \times C$

$N = 3 \times 1$ $\boxed{N=3}$

(A) 0.5 N
(D) 3 N
(C) 2 N

Q 84 The dissociation energy of $CH_4(g)$ is 360 Kcal mol^{-1} and that of $C_2H_6(g)$ is 620 Kcal mol^{-1}. The C-C bond energy is

(A) 260 Kcal mol^{-1}
(B) 180 Kcal mol^{-1}
(C) 130 Kcal mol^{-1}
(D) 80 Kcal mol^{-1}

$$\boxed{\begin{array}{r}620\\360\\\hline 260\end{array}}$$

130

$\frac{260}{2}$

Q 85 A triangle bipyramidal molecule has bond angle of

(A) 90^0
(B) $109^0 28'$
(C) 120^0
(D) $90^0, 120^0$

Q 86 An example of a salt dissolved in water to give acidic solution is

(A) Ammonium chloride
(B) Sodium acetate
(C) Potassium nitrate
(D) Barium bromide

Q 87 Maximum deviation from ideal gas is expected from

(A) H_2
(B) N_2
(C) CH_4
(D) NH_3

$n = \frac{22}{28}$

Q 88 Number of carbons atoms present in 22g CO_2 is

(A) 6.02×10^{23}
(B) 3.01×10^{23}
(C) 3.01×10^{-23}
(D) 6.02×10^{-23}

Q 89 Which of the following contains both ionic and covalent bonds?

(A) H_2O
(B) NaOH
(C) C_6H_5Cl
(D) NO_2

Q 90 Tyndall effect is more effectively shown by

(A) True solution
(B) Lyophillic colloid
(C) Lyophobic colloid
(D) Suspensions

6.02×10^{-23} 3.01×10^{23}

BIOLOGY (Q 91-135) TO BE ATTEMPTED BY CANDIDATES OF PCB & PCBM STREAMS

Q 91 An action potential in the nerve fibre is produced when positive and negative charges on the outside and the inside of the axon membrane are reversed because

(A) More potassium ions enter the axon as compared to sodium ions leaving it
(B) More sodium ions enter the axon as compared to potassium ions leaving it
(C) All potassium ions leave the axon
(D) All sodium ions enter the axon

Q 92 Sickle cell anaemia is:

(A) Autosomal dominant inheritence
(B) X-linked recessive inheritance
(C) Autosomal recessive inheritance
(D) X-linked dominant inheritance

Q 93 Match the followings from Column I and Column II

A. Hypothalamus	i) Relaxin
B. Anterior pituitary	ii) Estrogen
C. Testis	iii) FSH and LH
D. Ovary	iv) Androgens
	v) Gonadotropins releasing hormone

(A) A= v, B= iii, C= iv, D= ii
(B) A= v, B= iii, C= ii, D= iv

(C) A= i, B= ii, C= iv, D= iii (D) A= iii, B= v, C= iv, D= ii

Q94 Which one of the following sets includes pathogenic arthropods?
(A) Tsetse fly, mosquito, flea-plague
(B) Crab, *culex*, spider
(C) *Anopheles*, *culex*, cray fish
(D) Silver fish, house fly, sandfly

Q95 Length of petiole increases due to division of
(A) Apical meristem
(B) Lateral meristem
(C) Intercalary meristem
(D) Distal meristem

Q96 Which of the following is a bacterium involved in denitrification?
(A) *Nitrococcus*
(B) *Nitrosomonas*
(C) *Pseudomonas*
(D) *Nitrobacter*

Q97 Which is correctly matched?
(A) Apiculture : Honey bee
(B) Pisciculture : Silk moth
(C) Sericulture : Fish
(D) Aquaculture : Mosquitoes

Q98 In genetic engineering, the antibiotics are used
(A) As selectable markers
(B) To select healthy vectors
(C) As sequences from where replication starts
(D) To keep the cultures free of infection

Q99 A baby has been born with a small tail. It is the case exhibiting:
(A) Retrogressive evolution
(B) Mutation
(C) Atavism
(D) Metamorphosis

Q100 During sewage treatment, biogases are produced which include:
(A) Methane, hydrogen sulphide, carbon dioxide
(B) Methane, hydrogen sulphide, oxygen
(C) Methane, hydrogen sulphide, sulphur dioxide
(D) Methane, hydrogen sulphide, nitrogen

Q101 Reproductive isolation between segments of a single population is called:
(A) Sympatry
(B) Allopatry
(C) Population divergence
(D) Disruptive divergence

Q102 Lactobacillus mediated conversion of milk to curd results because of
(A) Coagulation and partial digestion of milk fats
(B) Coagulation and partial digestion of milk proteins
(C) Coagulation of milk proteins and complete digestion of milk fats
(D) Coagulation of milk fats and complete digestion of milk proteins

Q103 Carcinoma refers to:
(A) Malignant tumors of skin or mucous membrane
(B) Malignant tumor of color
(C) Benign tumor of connective tissue
(D) Malignant tumor of muscular tissue

Q104 Which one of the following processes during decomposition is correctly described?
(A) Fragmentation: carried out by organisms such as earthworm
(B) Humification: leads to accumulation of a dark coloured substance called humus which undergoes microbial action at a very fast rate
(C) Catabolism: last step in the decomposition under fully anaerobic condition
(D) Leaching: water soluble inorganic nutrients rise to top layers of soil

Q105 Photochemical smog formed in congested metropolitan cities mainly consists of:

(A) O_3, PAN and NO_x
(B) Smoke, PAN and SO_2
(C) Hydrocarbons, SO_2 and CO_2
(D) Hydrocarbons, O_3 and SO_2

Q 106 Reverse transcriptase is
(A) RNA dependent RNA polymerase
(B) DNA dependent RNA polymerase
(C) DNA dependent DNA polymerase
(D) RNA dependent DNA polymerase

Q 107 Which one of the following pairs of geographical areas show maximum biodiversity in our country?
(A) Sunderbans and Rann of Kutch
(B) Eastern Ghats and West Bengal
(C) Eastern Himalaya and Western Ghats
(D) Kerala and Punjab

Q 108 Identify the plant parts whose transverse section shows clear and prominent pith
(A) Dicot and monocot stems
(B) Dicot stem and monocot root
(C) Dicot and monocot roots
(D) Dicot stem and dicot root

Q 109 Bone marrow is made up of
(A) Muscle fibres and fatty tissue and blood vessels
(B) Fatty tissue and areolar tissue
(C) Fatty tissue, blood vessels and cartilage
(D) Fatty tissue, areolar tissue and blood vessels

Q 110 Which one of the following is not a live vaccine:
(A) BCG vaccine
(B) Cholera vaccine
(C) OPV
(D) Measles vaccine

Q 111 Important site for formation of glycoproteins and glycolipids is
(A) Vacuole
(B) Golgi apparatus
(C) Plastid
(D) Lysosome

Q 112 The female gametophyte in angiosperm is:
(A) Carpel
(B) Ovule
(C) Embryo sac
(D) Egg

Q 113 Which set is similar:
(A) Corpus luteum – Graafian follicle
(B) Sebum – Sweat
(C) Bundle of His – Pacemaker
(D) Vitamin B_7 – Niacin

Q 114 Which of the following statement is not correct:
(A) Pollen grains of many species can germinate on the stigma of the flower, but only one pollen tube of the same spore grows into the style.
(B) Insects that consume pollen or nectar without bringing about pollination are called pollen/nectar robbers.
(C) Pollen germination and pollen tube growth are regulated by chemical components of pollen interacting with those of pistil.
(D) Some reptiles have also been reported as pollinators in some plant species.

Q 115 Papillary muscles are found in mammalian
(A) Auricles of heart
(B) Ventricles of heart
(C) Pupil of eye
(D) Vallate papilla

Q 116 A Women with B positive blood group married a man with A positive blood group. The mother of both belongs to O negative blood group. What would be the probability of AB negative blood grop in their future off springs?
(A) 1/4
(B) 1/16

(C) 9/16 (D) 3/16

117 Choose the correct statement/statements regarding osmosis. In osmosis, the solvent molecules flow from the region of -

i. Lower DPD to the region of higher DPD through a semi permeable membrane(SPM).
ii. Higher water potential to the region of lower water potential through a SPM
iii. Solution with low solute concentration to a region of high solute concentration through a SPM.

(A) i
(B) i,ii
(C) ii,iii
(D) i, ii,iii

118 The source and the transgens used in developing Bt cotton is :

(A) *Agrobacterium tumifacien* →cry AC and AB
(B) *Bacillus thuringiensis* →cry IAB and cry IIAC
(C) *Bacillus thuringiensis* →cry IAC and cry 11AB
(D) *Agrobacterium tumifacien* →cry IAC and II AB

119 Bio-fortified wheat:

(A) Sonalika
(B) Sharbati Sonora
(C) Atlas 66
(D) Himgiri

120 The compound eyes of cockroach are adapted for :

(A) Diurnal mosaic vision with less resolution but more sensitivity
(B) Diurnal mosaic vision with more resolution but less sensitivity
(C) Nocturnal mosaic vision with less resolution but more sensitivity
(D) Nocturnal mosaic vision with more resolution but less sensitivity

121 Find the example of analogous organs:

(A) Thorns of pomegranate and spines of prickly pear
(B) Scale leaves of Onion and spines of prickly pear
(C) Tendrils of *cardiospermum* and bulbils of *Agave*
(D) Tendrils of *Vitis* and thorns of *Carissa*

122 Adnate anthers refer to :

(A) filaments attached to the base of anther
(B) filament attached along the whole length of anther
(C) filament attached to the back of anther
(D) anther lobes attached with the filament in the middle portion with both ends free

123 Choose the mismatch among the following:

(A) First movements of foetus and appearance of the hair on the head → during 5th month of pregnancy
(B) Development of limbs and digits → end of second month of pregnancy
(C) Formation of heart and its beating → after one month of pregnancy
(D) Separation of eye lids, formation of eyelashes → end of nine months of pregnancy

124 Sterlization techniques are generally full proof methods of contraception with the least side effects. Yet, this is the last option for the couple because:

i. It is almost irreversible.
ii. Of the misconception that it will reduce sexual urge.
iii. It is a surgical procedure.
iv. Of lack of sufficient facilities in many parts of the country.

Choose the correct answer

(A) (i) and (iii)
(B) (ii) and (iii)
(C) (ii) and (iv)
(D) (i), (ii), (iii) and (iv)

Q.125 Which particular process was used by M.Meselson and F.Stahl in order to study the semi-conservative replication of DNA:

(A) Centrifugation
(B) Chromatography
(C) Density gradient centrifugation
(D) Buoyant density centrifugation

Q.126 In *lac* operon, the genes a, I, y and z code respectively for:

(A) Transacetylase, repressor protein, permease, B-galactosidase
(B) Repressor protein, permease, B-galactosidase, transacetylase
(C) Transacetylase, permease, B-galactosidase, repressor protein
(D) Permease, transacetylase, repressor protein, B-galactosidase

Q.127 Shrunken appearance of erythrocytes is called

(A) Plasmolysis
(B) Incipient plasmolysis
(C) Deplasmolysis
(D) Crenation

Q.128 Darwin's finches are related to which of the following evidences?

(A) Fossils
(B) Embryological
(C) Anatomical
(D) Biogeographical

Q.129 *Archaeopteryx* is a missing link between:

(A) Pisces and Amphibia
(B) Reptiles and Aves
(C) Amphibia and Aves
(D) Reptiles and Mammals

Q.130 During which stage of meiosis, do the sister chromatids begin to move towards the poles?

(A) Prophase-I
(B) Telophase-I
(C) Anaphase-II
(D) Anaphase-I

Q.131 Transplantation of tissue/organs to save certain patients often fails due to rejection of such tissue/organs by the patient. Which type of immune response is responsible for such type of rejection:

(A) Auto-immune response
(B) Humoral immune response
(C) Physiological response
(D) Cell-mediated immune response

Q.132 A biologist studied the population of rats in a barn and observed that average natality was 250, average mortality 240, immigration 20 and emigration 30. The net increase in population was:

(A) 10
(B) 15
(C) 05
(D) Zero

Q.133 Cellulose is a polymer of

(A) Alpha-glucose
(B) Alpha-fructose
(C) Beta-glucose
(D) Beta-fructose

Q.134 Oral contraceptive pills help in birth control by:

(A) Killing sperms
(B) Killing ova
(C) Preventing ovulation
(D) Forming barrier between sperms and ova

Q.135 Class of enzymes contained in lysosomes is

(A) Lyases
(B) Ligases
(C) Hydrolases
(D) Transferases

N = 250
M = 240
a/c = 20
exit = 30

250
240

10
20